

Muskan Dhamuria

Mobile: (+65) 96787615 Email: muskandhamuria3k@gmail.com [Linkedin](#) [Portfolio](#)

EDUCATION

Singapore Management University

Aug 2024 – May 2028

Bachelor of Science Information Systems – Double Major in Product Development and Cybersecurity

- GPA: Cum Laude (Distinction)
- Activities : SMU Women in Tech , SMU .Hack , UWS STEMentorship 2025 (Mentee), SMU AI Club
- Awards: Most Feasible Solution in Hack For Cities Hackathon 2026, Honourable Mention Fintech Innovator's Hackathon 2026
- Teaching Assistant: Data Management

WORK EXPERIENCE

SP Group

Jul 2026 – Dec 2026

RPA Software Engineer

- Designed and implemented RPA scripts using .NET, JavaScript, and HTML, focusing on modular logic, error handling, and performance optimization.
- Engineered automation solutions using Automation Anywhere to replace manual workflows with scalable, code-driven processes.
- Integrated automation workflows with existing systems, debugged and resolved runtime issues, and contributed to improving system reliability and efficiency through structured testing and iteration.

Science Centre Singapore

Jan 2022

Student Intern

- Developed an interactive 'Young Engineer Trail' and online modules for the 'Young Digital Fabricator' badge program.
- Took lead presentation role during final delivery after unforeseen team absence.
- Took ownership of end-to-end delivery after teammate withdrawal, ensuring on-time project completion under increased workload.

PORTFOLIO

ESMOS Cloud-Native Healthcare Platform

- Designed and deployed cloud-native enterprise applications on Azure Kubernetes Service (AKS) using Docker, ArgoCD GitOps, and CI/CD pipelines. Implemented TLS-secured ingress routing with NGINX Ingress and cert-manager, container orchestration, persistent storage provisioning, monitoring/log analytics, and automated deployment workflows for scalable microservice-based healthcare systems.
- Technologies: Kubernetes, Docker, Azure AKS, Azure Container Registry, ArgoCD, GitHub Actions, NGINX Ingress, cert-manager, PostgreSQL, Azure Files, Azure Managed Disks

Questpass - Fair-access presale platform to combat ticket scalping

- Developed a presale platform where fans earn access through engagement-based quests (trivia, listening verification, mini-games), ensuring genuine fans get presale codes and artists build loyalty.
- Achieved Higher Distinction for the project during coursework.
- Technologies: Vue.js, Vue Router, Pinia, Bootstrap 5, Firebase (Auth, Firestore, Cloud Functions), Node.js, Spotify API, Gemini API, Jambase API, Google Maps API, EmailJS, Deezer API

Web Vulnerability Scanner

- Detects common security flaws such as SQL Injection, reflected XSS, and missing HTTP security headers. The tool uses a BFS-based crawler to explore web applications, test inputs in forms and parameters, and identify vulnerabilities with a risk-based scoring system (HIGH/MEDIUM/LOW). It includes features like rate limiting, stealth headers, and duplicate URL handling to simulate realistic scanning behavior, and generates detailed reports in JSON, HTML, and CSV formats for analysis.
- Technologies: Python, Requests & BeautifulSoup, Web Crawling (BFS), Application Security Testing (SQLi, XSS), Security Headers, Data Reporting (JSON, HTML, CSV)

Urbanix - AI-powered urban policy optimization platform

- Implemented multi-objective optimization logic and Pareto frontier analysis to simulate and evaluate urban policy bundles. Developed data ingestion and transformation pipelines for Excel-based urban datasets and real-time visualization dashboards.
- Awarded Most Feasible Solution - Hack For Cities Hackathon 2026

- Technologies: React.js, TypeScript, Tailwind CSS, Recharts, SheetJS, Node.js, Ollama, JavaScript

SKILLS

- **Programming & Backend:** Python, JavaScript, TypeScript, PHP, Bash, Flask, APIs, Firebase, Node.js, Java
- **Web & Frontend:** HTML, CSS, React, Vue, Bootstrap, Tailwind CSS
- **Cybersecurity & Operating Systems:** Kali Linux, Metasploitable, Linux System
- **Data & Analytics:** Power Query, Tableau, MySQL, Pandas, Matplotlib, Folium, Excel
- **Others:** Figma, Signavio, Azure, Docker